InFocus	SW ENGINEERING
CONTROLLED ISSUE INFORMATION	# INFRARED KEYCODE SPECIFICATION **009-1067-00**

Revision History					
Revision	**PCR #**	**Date**	**Initiator/Doc Control**	**Pages**	**Description**
A	P5137	06/02/06	M. Girardelli/J. Dixon		Initial Release
B	P5259	9/27/06	M. Girardelli/W. Glasscock	6,7	Discrete Blank Off and Discrete Blank On keycodes were added
C	P5663	6/24/08	A. Benenson	5,6,7	Added reserved codes 1, 5, 6, 21, 22, 31, and 56. Changed codes 9, 27-30 to reserved. Changed code 10 to help. Added new codes 69-79 for new aspect ratios and motorized zoom/focus/shift.
D	P5709	10/20/08	A. Benenson	6,7	'New Mute' (code 41) name changed to 'Mute' and revised description to be clearer. 'Discrete User Preset 2' and 'Discrete User Preset 3' (codes 65 & 66) descriptions revised; these codes access ISF day/night presets if projector supports ISF presets instead of user 2 & 3 presets.
E	P5745	12/18/08	A. Benenson	5,6	Changed the following codes that are not transmitted to N/A reserved codes: Light (15), Reset lamp hour counter (24), USB flash (42)
F	P5764	03/13/09	L.Hindman / P.Young	All	Changed Dept. to SW Engineering.
G	P5829	5/8/09	A. Benenson	6	Removed detail about USB behavior for next slide, previous slide (not needed for remote manufacturer)

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1. Purpose

This document will define the remote control keycodes that are supported on InFocus projectors.

2. Scope

This document is targeted at the engineering and OEM-inbound partners for the design of IR transmitters (e.g. remotes) and receivers (e.g. projectors).

3. Infrared Keycodes

The following table lists all of the currently designated infrared keycodes. Additional keycodes should be added to the end of this list. All unused values before the last specified keycode are reserved and should not be used.

InFocus Keycodes

		Code		
Key Name	**Description**	**Decimal**	**Hexadecimal**	**Binary**
N/A	Reserved (selects computer 1 source)	1	01h	0000 0001
Menu	Toggles menu state up or down	2	02h	0000 0010
Brightness Plus	Increases brightness	3	03h	0000 0011
Volume Minus	Decreases volume	4	04h	0000 0100
N/A	Reserved (selects video 1 source)	5	05h	0000 0101
N/A	Reserved (selects scribble mode)	6	06h	0000 0110
Brightness Minus	Decreases brightness	7	07h	0000 0111
Volume Plus	Increases volume	8	08h	0000 1000
N/A	Reserved (external media player)	9	09h	0000 1001
Help	Displays help menu	10	0Ah	0000 1010
Right Mouse Button	Sends right mouse click	11	0Bh	0000 1011
Left Mouse Button	Sends left mouse click	12	0Ch	0000 1100
Standby	Toggles standby state on or off	13	0Dh	0000 1101
Freeze	Toggles freeze state on or off	14	0Eh	0000 1110
N/A	Reserved (enable backlighting on remote control, not transmitted)	15	0Fh	0000 1111
Source Select	Selects next available source	16	10h	0001 0000
Cursor Left	Moves menu cursor left	17	11h	0001 0001
Cursor Right	Moves menu cursor right	18	12h	0001 0010
Cursor Up	Moves menu cursor up	19	13h	0001 0011
Cursor Down	Moves menu cursor down	20	14h	0001 0100
N/A	Reserved (selects computer 2 source)	21	15h	0001 0101
N/A	Reserved (selects video 2 source)	22	16h	0001 0110
Power	Toggles power state on or off	23	17h	0001 0111
N/A	Reserved (reset lamp hour counter, not transmitted)	24	18h	0001 1000

Computer Toggle	Selects next available computer source	25	19h	0001 1001
Video Toggle	Selects next available video source	26	1Ah	0001 1010
N/A	Reserved (selects computer 4 source)	27	1Bh	0001 1011
N/A	Reserved (selects computer 5 source)	28	1Ch	0001 1100
N/A	Reserved (selects video 3 source)	29	1Dh	0001 1101
N/A	Reserved (selects computer network A)	30	1Eh	0001 1110
N/A	Reserved (selects computer network B)	31	1Fh	0001 1111
Keystone Plus	Increases keystone	32	20h	0010 0000
Keystone Minus	Decreases keystone	33	21h	0010 0001
N/A	Reserved (increment selected setting)	34	22h	0010 0010
N/A	Reserved (decrement selected setting)	35	23h	0010 0011
Next Slide	Moves to next slide in powerpoint	36	24h	0010 0100
Previous Slide	Moves to previous slide in powerpoint	37	25h	0010 0101
Zoom Plus	Increases zoom ratio	38	26h	0010 0110
Zoom Minus	Decreases zoom ratio	39	27h	0010 0111
Zoom Toggle	Toggles zoom state on or off	40	28h	0010 1000
Mute	Toggles audio mute state on or off	41	29h	0010 1001
N/A	Reserved (enable USB flash mode, not transmitted)	42	2Ah	0010 1010
Custom	Performs programmed action of Custom Key	43	2Bh	0010 1011
Presets	Toggles through available presets	44	2Ch	0010 1100
Source1	Selects programmed source	45	2Dh	0010 1101
Source2	Selects programmed source	46	2Eh	0010 1110
Source3	Selects programmed source	47	2Fh	0010 1111
Source4	Selects programmed source	48	30h	0011 0000
Blank	Toggles state of blank screen on or off	49	31h	0011 0001
Select	Selects current menu item	50	32h	0011 0010
Auto Image	Activates automatic image re-acquisition	51	33h	0011 0011
Resize	Toggles through available resize modes	52	34h	0011 0100
Overscan	Toggles overscan state on or off	53	35h	0011 0101
Contrast Plus	Increases contrast	54	36h	0011 0110
Contrast Minus	Decreases contrast	55	37h	0011 0111
N/A	Reserved (aspect ratio toggle)	56	38h	0011 1000
Discrete Power Off	Commands the system to power down	57	39h	0011 1001
Discrete Power On	Commands the system to power up	58	3Ah	0011 1010
Discrete Resize Native	Selects the Native aspect ratio	59	3Bh	0011 1011

Discrete Resize 16:9	Selects the 16:9 aspect ratio	60	3Ch	0011 1100
Discrete Resize 4:3	Selects the 4:3 aspect ratio	61	3Dh	0011 1101
Discrete Resize Letterbox	Selects the Letterbox aspect ratio	62	3Eh	0011 1110
Discrete Resize Natural Wide/Panorama	Selects the Natural Wide/Panorama aspect ratio	63	3Fh	0011 1111
Discrete User Preset 1	Selects the User Preset 1	64	40h	0100 0000
Discrete User Preset 2	Selects the User Preset 2 (or selects ISF Day preset if projector supports that instead of user preset 2)	65	41h	0100 0001
Discrete User Preset 3	Selects the User Preset 3 (or selects ISF Night preset if projector supports that instead of user preset 3)	66	42h	0100 0010
Discrete Blank Off	Turns off the blank screen	67	43h	0100 0011
Discrete Blank On	Turns on the blank screen	68	44h	0100 0100
Discrete Resize Auto	Selects the Auto aspect ratio	69	45h	0100 0101
Discrete Resize 15:9	Selects the 15:9 aspect ratio	70	46h	0100 0110
Discrete Resize 16:10	Selects the 16:10 aspect ratio	71	47h	0100 0111
Focus Near	Motorized focus nearer	72	48h	0100 1000
Focus Far	Motorized focus further	73	49h	0100 1001
Zoom In	Motorized zoom in	74	4Ah	0100 1010
Zoom Out	Motorized zoom out	75	4Bh	0100 1011
Shift Left	Motorized lens shift move left	76	4Ch	0100 1100
Shift Right	Motorized lens shift move right	77	4Dh	0100 1101
Shift Up	Motorized lens shift move up	78	4Eh	0100 1110
Shift Down	Motorized lens shift move down	79	4Fh	0100 1111